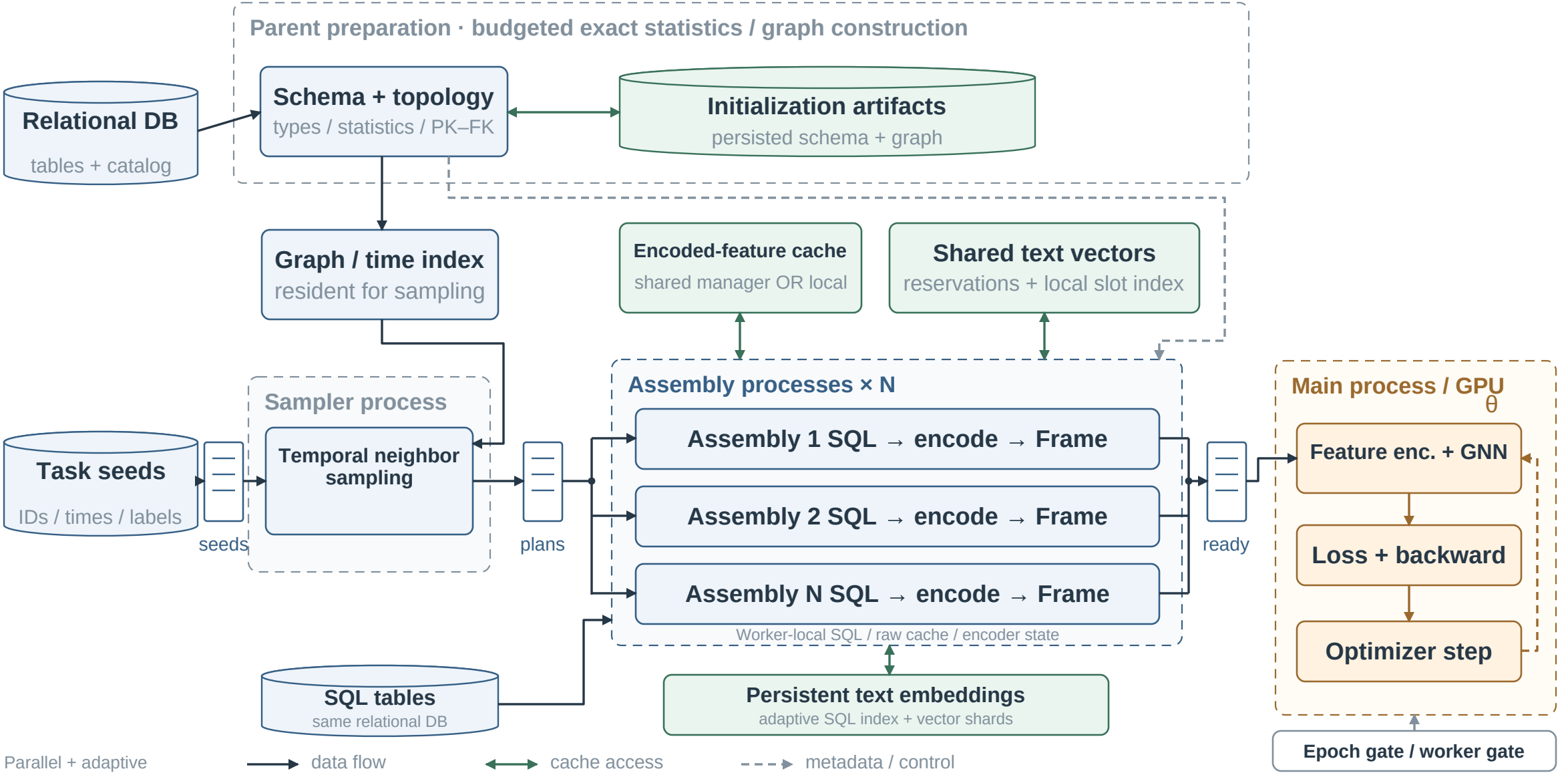
Parent preparation · budgeted exact statistics / graph construction
Relational DB
tables + catalog
Schema + topology
types / statistics / PK–FK
Initialization artifacts
persisted schema + graph
Graph / time index
resident for sampling
Encoded-feature cache
shared manager OR local
Shared text vectors
reservations + local slot index
Sampler process
Task seeds
IDs / times / labels
seeds
Temporal neighbor sampling
plans
Assembly processes × N
Assembly 1 SQL → encode → Frame
Assembly 2 SQL → encode → Frame
Assembly N SQL → encode → Frame
Worker-local SQL / raw cache / encoder state
ready
Main process / GPU
θ
Feature enc. + GNN
Loss + backward
Optimizer step
SQL tables
same relational DB
Persistent text embeddings
adaptive SQL index + vector shards
Epoch gate / worker gate
Parallel + adaptive
data flow
cache access
metadata / control